

Group: PBL 11 Jade
The University of Queensland
Saint Lucia, QLD, 4072

May 28th, 2021
Australian Aviation Rescue Fire Fighting Service
Brisbane, Queensland
Australia

Dear Sir or Madame,

Attached is a copy of the finalised report with all details surrounding the unmanned firefighting vehicle. The main findings of the report are:

- The best design to use for the vehicle prototype, focusing on the four main subsystem: control, fluid delivery, structural and drive.

Presented below are all initial design ideas revolving around a selection of criteria, and the process of finalising these ideas in order to produce the most effective system. Also included below are the overviews of each subsystem with detailed diagrams, risk analysis, scale up considerations, financial details, an insightful conclusion of the design experiences and a reflection of the project process.

The prototype consists of aluminium and acrylic for the main structure of the system as these materials are most suitable for the criteria. The aluminium was low cost while providing a stable structure and acrylic was utilized for the internal support, as well as protection from any water leakage. Both materials are light weight and easy to alter appropriately to suit the vehicle. Two servomechanisms, a small plastic nozzle and 1 litre container are utilized within the fluid delivery subsystem, while two dc motors and wheels are utilized within the drive subsystem. These subsystems are controlled by a PCB and Arduino which are both utilized within the control subsystem. All components were justified based on weight, cost, dimensions, as well as specific criteria for the individual components.

Yours sincerely,
PBL 11 Jade

[Redacted Signature]

Kurt Armamento

[Redacted Address Line 1]

[Redacted Address Line 2]

[Redacted Address Line 3]

[Redacted Address Line 4]

UNIVERSITY OF QUEENSLAND

Faculty of Engineering, Architecture and Information Technology

Bachelor of Engineering

ENGG1100

FINAL REPORT:
Unmanned Firefighting Vehicle

By

[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]
KURT ARMAMENTO	[REDACTED]
[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]

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Semester 1

Executive Summary

Mitigating the immediate danger aviation firefighters are placed in requires the development of unmanned firefighting technologies. *The Australian Aviation Rescue Fire Fighting Service* (ARFF) have commissioned the development of an unmanned firefighting vehicle which it can be remotely controlled and driven in close vicinity of emergency locations to extinguish the fires.

The unmanned firefighting vehicle prototype was designed with the following requirements:

- The vehicle must be able to extinguish six fires (heat sources) placed within the test space
- The vehicle must be able to travel steadily forwards and backwards
- The vehicle must comprise of an apparatus to pressurise and direct firefighting fluid
- There must be an automated, semi-automated or manually controlled mechanism in place to remotely control the vehicle's drive and fluid delivery
- The vehicle's structure must contain the drive, fluid delivery and control mechanisms, and must be robust to resist movement during operation.

Following the investigation of existing designs, the final design was constructed consisting of the following components:

Structural Subsystem:

- Two separated acrylic platforms, with holes in the surfaces, suspended apart by stainless steel bolts.

Fluid Delivery Subsystem:

- Mini Submersible Pump DC 3V-5V connected to a nozzle on two servomechanisms for both horizontal and vertical positioning.

Control Subsystem:

- An Arduino and L293D Motor Driver IC powered by eight AA 1.5V batteries, and controlled using an IR remote and receiver combination.

Drive Subsystem:

- Rear wheel drive using DC Geared 10V motors and lightweight wheels at the front

Altogether, the proposed design satisfied the design criteria with intuitive control mechanisms combined and precision-based fluid delivery system. The unmanned fire fighting vehicle was able to succeed in extinguishing targeted fires during test, validating the final design. Due to the small-scale vehicle's success and low-cost design, a scale up and mass deployment of unmanned fire fighting vehicles is recommended.

There are a number of areas to improve on the design, which have been recommended for further study. Such areas include investigation into other structure-based materials to better suit the full-scale model, investigation into alternate energy sources that prove to be more energy dense and experimentation surrounding different forms of nozzles to determine idealised flow rate. Additionally, the small-scale unmanned fire fighting vehicle is not suitable to be scaled up in its current state. There are numerous considerations which must be made before a full-scale prototype is constructed.

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1. Introduction

1.1 Context

In the event of aircraft emergencies at airports where fires are ignited, there are numerous challenges which need to be overcome by firefighters in order to extinguish the fires safely. However, growing complexity in aircrafts' onboard devices, composite construction and potentially hazardous cargo present increased difficulty when facing an emergency situation. It is thought that avoiding placing aviation firefighters in immediate danger will importantly mitigate the risk of loss of life.

1.2 Stakeholders

The Australian Aviation Rescue Fire Fighting Service (ARFF) commissioned the development of an unmanned firefighting vehicle with the first full scale prototype to be initially deployed at the Brisbane Airport.

1.3 Aim

This concept report aims to address the removal of aviation firefighters from immediate danger, specifically through the creation of a small-scale prototype of an unmanned firefighting vehicle. The proposed technology is intended to be centrally located amongst runway infrastructure, from which it can be remotely controlled and driven in close vicinity of the emergency location and extinguish the fires. The creation of the small-scale prototype was achieved by defining and scoping the problem, examining existing designs and drawing a conclusion about the best solution.

1.3.1 Objectives

This concept report aims to address the following objectives:

- To design a proof-of-concept prototype of an unmanned firefighting vehicle which can be remotely controlled to approach and extinguish fires at a close proximity.
- The vehicle must be able to extinguish six fires (heat sources) placed in various locations within the test space.
- The vehicle must be able to travel steadily forwards and backwards in a straight line.
- The vehicle must comprise of an apparatus to pressurise and direct firefighting fluid.
- There must be a mechanism in place to remotely control the vehicle's drive and fluid delivery, whether it be automated, semi-automated or manually controlled.
- The vehicle's structure must contain the drive, fluid delivery and control mechanisms, and must be robust to resist movement during operation.

1.3.2 Constraints

The project is based on the following constraints:

- The proof-of-concept prototype must fit within the box dimensions of 300mm x 200mm x 300mm with a maximum total mass of 3 kg.
- The vehicle must contain the drive, fluid delivery and structural subsystems within its structure (batteries and control mechanisms can be located externally).
- The power supplied by batteries must not exceed 13 V and must suffice for one 6-minute operation.
- The total cost of the proof-of-concept prototype must not exceed the AUD\$150 budget.
- The manufactured component and provided PCB must be incorporated into the vehicle.
- Only 1 L of water at ambient room temperature can be contained within the vehicle.

1.3.3 Assumptions

Several assumptions have been made that are carried throughout the design process. They are as follows:

- The locations fires (heat sensors) are already known, as determined in the test space specifications (*Appendix 1*).
- The automated firefighting vehicle is required to decrease the temperature heat sensors from 150°C to 100°C for the fire to be considered ‘extinguished’.
- A manufactured component will be created out of aluminium, with dimensions and specifications determined in the manufactured component dimensions (*Appendix 2*).
- The budget for the proof-of-concept prototype is sufficient to fund the project.
- The automated firefighting vehicle with operate with sufficient efficiency to replace firefighting officers.
- Success in the small-scale demonstration will equate to a success of the full-scale prototype to be used at the Brisbane Airport.

1.3.4 Project Scope

Table 1: Project Scope

In Scope	Out of Scope
Comprehensive engineering design of subsystem	Construction of the full-scale prototype
Sourcing of materials	Performance in varying terrain
Estimated bill of materials management	Efficiency in comparison to human firefighters
Construction of a working small-scale prototype	Training for use of the vehicle
Scale up consideration	Evaluation of different firefighting fluids
Graphical representation of final design	Logistics of implementation at airport
Small-scale prototype testing	Repairing the vehicle in case of failure
Detailed calculations for technical details	Ongoing maintenance of the vehicle
Adhering to design constraints such as dimensions, mass and cost	

1.4 Design Criteria

Comprehensive design criteria are necessary to evaluate both the subsystem and the system as whole, and determine how well each design option achieves task objectives and aligns with the constraints outlined in the brief. For the structural subsystem, the following criteria were used:

Table 2: Outline of Design Criteria

System Criteria	
Criteria A: Cost	This system criterion addresses the project constraint of constructing the proof-of-concept prototype within the AUD\$150 budget.
Criteria B: Size and Form	As outlined in the project brief, this criterion correlates to the system design option’s ability to remain within the dimensional requirements.
Criteria C: Mass	This refers to the total weight of the subsystem’s elements, and is in direct response to the weight constraints outlined in the project brief.
Subsystem Criteria	
Criteria D: Stability	This criterion relates specifically to the robustness of the structural subsystem, and correlates to the structural objectives of the project.

Criteria E: Sustainability	Success in this criterion refers to the environmental sustainability and recyclability of the given design option.
Criteria F: Heat Resistance	Heat resistance as a criterion refers to the ability for the structural subsystem to withstand and perform in heated environments.
Criteria G: Longevity	This refers to the expected life of subsystem components, and the ability for them to be maintained over extended periods of time.
Criteria H: Water Pressure	Water pressure refers to the fluid delivery subsystem's ability to project water at extended distances, relating specifically to the pumps water pressure.
Criteria I: Rotation of Nozzle	This criterion relates to a fluid delivery subsystem component's ability to adjust the position of the nozzle to target fires in varying locations.
Criteria J: Water Efficiency	Water efficiency is the ability of the vehicle to extinguish fires using a limited volume of water
Criteria K: Voltage/Current	This criterion judges how well the batteries meet the minimum required voltage and current for the whole system.
Criteria L: Accessibility	This criterion refers to the ability to use the batteries in the control subsystem and if there are available accessories for them.
Criteria M: Power	Power refers to the ability of the drive subsystem components to effectively operate under the load of the vehicle.

2. Technical Background

As part of the process to creating the final design, existing design options were analysed to get a better understanding of what makes an unmanned firefighting vehicle's design successful. Features of each design were investigated and concluded how they could've been incorporated into the small-scale prototype:

Table 3: Technical Background

Option	Photo	Features	Suitability for the Project
2.2.1 Fire Ox Firefighting Robot (Gabbart, 2015)	 <i>Figure: Fire Ox Firefighting Robot</i>	• Compact structure contains all subsystems within an impact resistance shell • Drives with six wheels, for even distribution of weight and stability in rough terrain environments • Possesses both water misting and directional projection to extinguish fires in both close and far range • Use of materials result in an extensively heavy vehicle	The <i>Fire Ox</i> Firefighting Robot was considered an exemplified full-scale unmanned firefighting vehicle. The key points from its design included the heat and impact force resistance to perform in a variety of extreme environments. However, its complex industrial design exceeded the needs of the small-scale prototype.

2.2.2 Cobham tEODor Bomb Disposal Robot (How It Works, 2013)	 <i>Figure: Cobham tEODor Bomb Disposal Robot</i>	• The control and drive subsystem are contained at the durable blast resistant base of a vehicle • The drive system consists of tracks rather than wheels. This provides stability and full contact with the ground • Robotic arm is positioned at front of the vehicle and has 360° horizontal rotation and 180° vertical rotation • Vehicle possesses a camera on robotic arm for remote use in unfamiliar environments	While the <i>Cobham tEODor</i> Bomb Disposal Robot serves a different function to a firefighting vehicle, its robust structural form and sophisticated robotic control was still applicable to this project. The robotic arm was considered to be utilised for the fluid delivery system, with the ability to target fires with precision.
2.2.3 WVF-5 Firefighting Robot https://firesafety.tips/fire-fighting-robots/	 <i>Figure: WVF-5 Firefighting Robot</i>	• Fluid delivery system is highly elevated to project fluid downward on the target • Impact shield at the front of the vehicle to protect control and drive systems • Emphasis of design is on dynamic fluid delivery with stable and robust base of vehicle • Tracks are used rather than wheels for drive subsystem • Intumescent materials used within the body of the robot to protect the control subsystem from high temperatures	The <i>WVF-5</i> Firefighting Robot is another unique design solution to unmanned firefighting. The key takeaways from the design included the importance of an elevated fluid delivery, low centre of mass and use of robust materials. Additionally, the specific use of materials provides protection to fragile components from extreme heat conditions and impact forces.

Although these existing designs are exemplified vehicles for the full-scale project, each design possess features which are applicable when developing the final design for the small-scale prototype. The knowledge gained assisted when making decisions about which components would be most effective for the final unmanned firefighting vehicle design.

3. Design Options

3.1 Functional Layout and Decomposition

The proof-of-concept prototype can be encompassed into seven main functions, as seen below in (Figure 1):

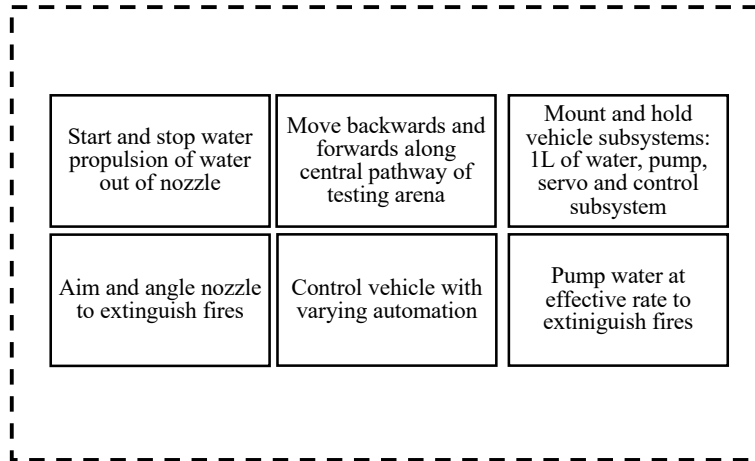


Figure 1: Functional Layout

In order to achieve such functions, the vehicle was separated into four distinct subsystems: drive, fluid delivery, structural and control. The specific elements of each subsystem can be seen below in the functional decomposition (Figure 2).

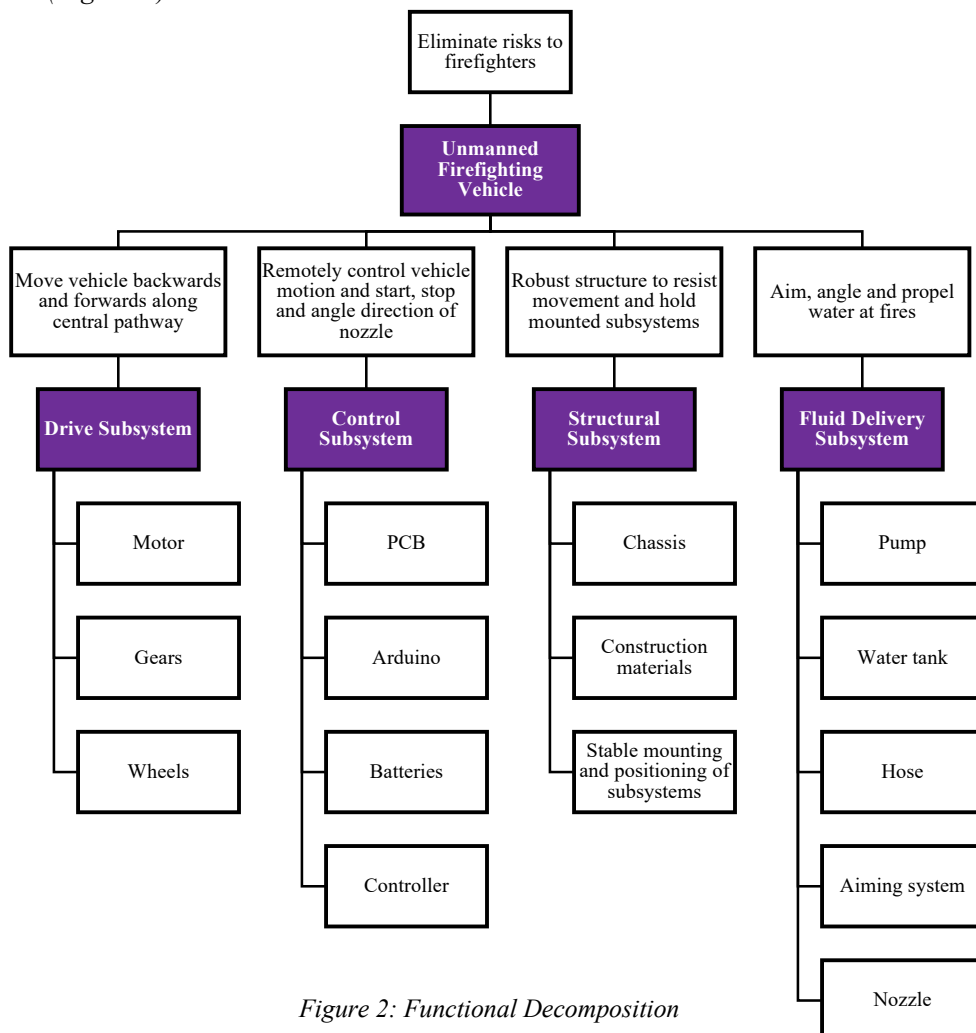


Figure 2: Functional Decomposition

3.1.1 Overview of Subsystems

In review of the overall system of the unmanned firefighting vehicle, each subsystem must achieve its own primary functions outlined above in order for the operation to be completed. The drive subsystem initiates the base movement of the vehicle, including maintaining a smooth acceleration and deceleration along the central pathway of the testing arena. The control subsystem is responsible for maintaining the control and powering of the drive and fluid delivery electrical components. The control subsystem is central in the manoeuvrability of the servo motors particularly the elevation and angling of the nozzle to aim at the targets. The structural subsystem aims to utilise robust materials to construct a stable structure that appropriately positions all vehicle components. The fluid delivery subsystem consists of a tank capable of holding 1 litre, a hose, nozzle and water pump that projects water at an efficient rate.

3.2 Design Process for Chosen Design

Every component of each subsystem was evaluated based on both system and subsystem specific criteria, by allocating a ranking from 1 to 10 (with 10 being the best). This was in order to draw a logical and justified conclusion about which option(s) would be best suited for the proof-of-concept prototype.

The final design for the structural subsystem explored features from existing designs to create one comprehensive design which best addressed the design criteria. The decision-making matrix, seen below in *Table 4*, evaluated the *Space Frame* and *Platform Chassis* (see *Appendix 1*), to make a justified decision about what structural design would be used for the small-scale prototype.

Table 4: Structural Subsystem Decision Making Matrix

Criteria	System Criteria			Subsystem Criteria				Total
	Criteria A: Cost	Criteria B: Size & Form	Criteria C: Mass	Criteria D: Stability	Criteria E: Sustainability	Criteria F: Heat Resist	Criteria G: Longevity	
Weighting	0.190	0.285	0.143	0.238	0.001	0.048	0.095	
Solution 1: Space Frame	3	3	2	3	2	2	2	2.713
Solution 2: Platform Chassis	5	4	5	2	3	1	5	3.807

**see Appendix 2 for weighting calculation in piecewise comparison matrix*

The *Space Frame* was considered most applicable due to the simplistic designs' low cost, low mass and size, however, did not achieve a sufficient score for stability to be used in its entirety.

Taking inspiration from the platform design, the final design was chosen to consist of two separated platforms rather than one, with holes in the surfaces, suspended apart in two levels. Adding another level to the platform chassis design improves the stability of the structure, offers more space for components to be positioned and provides protection for fragile components between the two levels. Each platform also possessed a small number of holes drilled, incorporated for the ease of wiring and plumbing of subsystems across platforms and providing a method to tether the various components to the structure.

Similarly, the fluid delivery subsystem explored numerous designs to create the final product. *Table 5* below shows the evaluation of the considered options for the rotation of the nozzle in order to determine which option would be used in the final design:

Table 5: Fluid Delivery Subsystem Decision Making Matrix

Criteria	System Criteria			Subsystem Criteria			Total
	Criteria A: Cost	Criteria B: Size & Form	Criteria C: Mass	Criteria J: Water Efficiency	Criteria I: Rotation of Nozzle	Criteria J: Water Pressure	
Weighting	0.13	0.20	0.07	0.20	0.33	0.07	
Solution 1: 2 x Servomechanisms	3	2	4	4	5	2	3.66
Solution 2: 2 x Gears in equilibrium and in contact	2	3	4	3	3	2	2.87

**see Appendix 3 for weighting calculation in piecewise comparison matrix*

It can be seen that *Solution 1* was the best option for the rotation of the nozzle. This solution consisted of a closed-loop mechanism that incorporates positional feedback in order to control the rotational or linear speed and position (Gastreich, 2018). Using horizontal and vertical servo motors in order to rotate the nozzle was considered to be most effective. The second solution, consisting of two gears in equilibrium and in contact with each other, would succeed however not as effectively as Option 1. The final design of the subsystem also consisted of a Mini Submersible Pump DC 3V-5V which has a water flow of 100L/H, a 5mm diameter Clear Vinyl Tubing in order to transport the water from the tank to the nozzle and a 1 litre clear plastic container to hold the water. These options were all chosen due to their ability to address the most important design criteria of being light weight and low cost.

The control subsystem also was influenced by existing projects in the Arduino community which fulfilled tasks similar to that which the unmanned firefighting vehicle must fulfill, such as driving forwards and backwards. By referencing existing projects and learning from issues which others have faced, components could be evaluated to determine which power source was most suitable for the final design.

Table 6: Control Subsystem Decision Making Matrix

Criteria	System Criteria			Subsystem Criteria			Total
	Criteria A: Cost	Criteria B: Size & Form	Criteria C: Mass	Criteria G: Longevity	Criteria K: Voltage/Current	Criteria L: Accessibility	
Weighting	0.067	0.267	0.067	0.267	0.267	0.067	
Solution 1: AA 1.5V Alkaline	5	5	5	4	3	4	4.142
Solution 2: 9V Alkaline	3	3	3	2	2	3	2.472

**see Appendix 4 for weighting calculation in piecewise comparison matrix*

Initially, 9-volt batteries were chosen as the main power source for the prototype. However, after researching existing projects utilising these batteries, it was found that 9-volt batteries do not supply sufficient current to power DC motors. Following the evaluation seen in *Table 6*, it was decided that the most appropriate battery was the AA 1.5V Alkaline Batteries. Overall, AA batteries are most cost, size and mass effective. It is the second smallest battery and has numerous battery holder options, but AA batteries only have a voltage of 1.5V which is not sufficient to power the whole system. This was overcome by using multiple AA batteries in series to increase the overall voltage. Control was also responsible for the method for which external input was received

and the supplied Arduino microcontroller. It was concluded that the most effective options for the power source and external input were a set of 8 AA 1.5V batteries, and an IR remote and receiver combination.

For the drive subsystem, it had to be evaluated which method of ground propulsion would be most efficient, as well as what component to use to achieve it.

Table 7: Drive Method Decision Making Matrix

Criteria	System Criteria			Subsystem Criteria			Total
	Criteria A: Cost	Criteria B: Size & Form	Criteria C: Mass	Criteria D: Durability	Criteria E: Sustainability	Criteria M: Power	
Weighting	0.14	0.05	0.14	0.24	0.05	0.19	
Solution 1: Wheeled	4	3	3	3	2	4	2.71
Solution 2: Continuous Tracks	2	4	2	4	3	3	2.44

**see Appendix 5 for weighting calculation in piecewise comparison matrix*

Of the two forms of ground propulsion evaluated above, a wheeled solution ranks higher than a tracked solution with a 2.71 compared to a 2.44 respectively. The increased movement ability of solution 1 allows the prototype to more efficiently move back and forth on the testing arena. Solution 1 proved to be more desirable mainly due to a lower cost, lower mass, and the increased moveability. Although solution 2 was a smaller size, possessed a higher durability and a better sustainability, the majority of these criteria had a lower weighting due to their secondary nature to the task of the drive subsystem. Whilst solution 2 had a higher durability, which is weighted very highly, it was deemed that the difference between the durability between the two solutions was not great enough to make a significant difference.

Table 8: Drive Component Decision Making Matrix

Criteria	System Criteria			Subsystem Criteria			Total
	Criteria A: Cost	Criteria B: Size & Form	Criteria C: Mass	Criteria D: Durability	Criteria E: Sustainability	Criteria M: Power	
Weighting	0.14	0.05	0.14	0.24	0.05	0.19	
Solution 1: 2 x Shaft universal 3V Motor / Gearbox Set	4	4	4	2	3	1	2.14
Solution 2: 2 x DC Geared 10V Motor	4	3	3	4	2	5	3.14

**see Appendix 5 for weighting calculation in piecewise comparison matrix*

Of the two brushless motors evaluated above, the 10V DC Motor ranked higher at 3.14, over the 3V Motor due to its increased power capabilities and increased durability. The increased power of *Solution 2*, due to its higher voltage rating, ensured that the motor could reliably move the vehicle's load down the central pathway of the testing arena. *Solution 2* ranked higher largely due to the motors' ability to address criterion of the project, as it was an appropriate price, made of sustainable materials and more durable than *Solution 1*. Durability was a key criterion as the motor was required to move the load smoothly along the pathway, thus requiring an appropriate voltage rating. Although *Solution 1* was lighter and smaller in size in comparison to *Solution 2*, power and durability were weighted higher as these criteria are key in fulfilling the objectives of the project, making the DC Geared 10V Motor the chosen drive component for the final design.

4. Design Analysis

4.1 Technical Design Details

Table 9: Scientific Analysis and Specifications for Subsystems

Subsystem	Specifications of Subsystem
Structural	The structural subsystem consisted of two 3mm acrylic sheets, cut to be 250mm x 150mm. The original sheet of acrylic was purchased for \$12.99, and only half of the sheet was utilised. Acrylic was chosen for the platforms of the vehicle, as with a Young's Modulus of 3.2 GPa (The Engineering ToolBox, 2003), a 250mm x 150mm sheet will only experience bending deflection of 1.77mm when placed under a 1 kg mass (see <i>Appendix 6</i>), and experience no elastic deformation. Additionally, as the vehicle's structure was perfectly symmetrical, the theoretical centre of mass would be in the centre of the vehicle. With a low centre of gravity, adding the numerous components to the vehicle's structure would not unbalance the vehicle in any way. In order to suspend the two platforms separately, a threaded aluminium rod was purchased for \$5.95 and cut into segments 10 cm in length. These were placed at the four corners of each platform and secured using 1/4-inch zinc nuts and washers. Placing the pillars at the corners ensured that the mass of the top platform was evenly distributed on the bottom platform, minimising the bending moment experienced by the lower platform.
Fluid Delivery	The fluid delivery subsystem encompasses the transportation of the water from the unmanned firefighting vehicle to the heat sources. The most suitable method for the transportation of 1 litre of water is using a rectangular container. The water is delivered to all heat sources using a pump, with a water pressure of 100L/H. Connecting to this pump is a 5mm diameter tube with a nozzle attached in order to spray the water in a controlled way. Two servomotors one placed upside down and one placed on its side to direct the water in specific directions. These components are a suitable size as it fits the dimension constraints and is minimal weight. The overall budget for this subsystem is \$45.76.
Control	The control subsystem was responsible for the technical aspects of the UFV. This included powering the vehicle, receiving external input and controlling the drive and the fluid delivery subsystems. Control was responsible for three of the components used in the final prototype, the power source, the method for which external input is received as well as the supplied Arduino microcontroller. It was concluded that the most effective options for the power source and external input were a set of 8 AA 1.5V batteries, and an IR remote and receiver combination. The power source has dimensions of 58mm x 60mm x 30mm using an 8 AA battery holder. In total, it supplied 12 V to the whole system and costs a total of AUD\$8.25. The IR receiver was an approximately 5mm cube with a read distance of up to approximately 20 metres and a centre frequency of 36 kHz. The approximate read characteristics were 45 degrees horizontal and vertical. The used IR remote had dimensions of 40mm x 10mm x 85mm with 21 buttons. The IR combination had a total cost of AUD\$12.90. The Arduino used was the Uno rev3. It had dimensions of 75mm x 13mm x 53mm and costs AUD\$29.95. A L293D Motor Driver IC was also used to provide high power output to control the Drive subsystem. It was a H-bridge with 8 pins on either side of the chip and costs AUD\$4.95. The control system had a total cost of AUD\$56.05.

Drive	The drive subsystem is responsible for the movement of the unmanned firefighting vehicle forwards and backwards. The drive system contains three central parts which play a pivotal role in the functionality of the UFV. It was deemed that the most efficient method of ground propulsion would be the use of wheels connected to motors to drive them. The drive subsystem features a rear-wheel-drive configuration powered by two individually powered wheels, each receiving 5V from the batteries. This layout allows the vehicle to remain efficient whilst lowering the cost compared to a four-wheel-drive system. The 10V DC Geared Motor with Rubber Wheels were utilised due to their suitability for Arduino based vehicle applications and power capabilities. Combined with a low 48:1 gear ratio on each wheel, ample amounts of both torque and speed are provided to the wheels. The 48:1 gear ratio shows that the motor is high speed and low torque, which is preferred as high torque due to relatively light weight load on motors. To balance the vehicle, a set of unpowered light weight wheels were placed at the front. The total cost of the two powered wheels came out as \$19.90, and the unpowered wheels and casing came out to \$18.43. In total, the total drive subsystem had a cost of \$38.33.
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4.1.1 Materials Selection

For the construction of the entire subsystem, specific consideration was taken in selecting appropriate materials for each subsystem. In order to ensure the optimal material choice was made, in terms of addressing design criteria such as mass, durability and sustainability, each material's advantages and disadvantages were identified:

Table 10: Material Design Options

Material	Advantages	Disadvantages
3.1.2.1 Aluminium	- Aluminium has a low density of 2.70 g/cm³ - Resistant to corrosion - Recyclable without loss of quality	- Lower tensile strength (90 MPa) than other metals (AZO Materials, 2005) - High thermal conductivity
	Summary: Aluminium is a lightweight, durable and easily manufacturable metal. Although it is not the strongest of metals explored, for the purpose of the proof-of-concept prototype, it will be sufficiently robust to stably contain subsystems and resist small forces.	
3.1.2.2 Steel	- Thermal, chemical or mechanical resistance - High durability - High capability to absorb impact energy	- Strength is reduced when exposed to high temperatures - Susceptible to corrosion - Higher density than other metals (7.87 g/cm³)
	Summary: Steel is the heaviest and strongest of all materials investigated. While its robustness and thermal resistance would position it as the prime choice for the full-scale prototype, its mass density would likely exceed the required weight outlined in the brief.	
3.1.2.3 Wood	- Cheapest of all materials - Density of 0.50-0.90 g/cm³ is less than most metals (Mitchell, 2018) - Numerous types of wood to suit various project needs - High tensile strength (100 MPa)	- Combustible material - Wood is hygroscopic so will swell and shrink when wet
	Summary: Wood is an inexpensive, lightweight and robust material. However, wood fails to be considered based on the fact that it is combustible and hygroscopic. It is not appropriate for a firefighting vehicle, which is exposed to both extreme heat and water, to be made of such a material.	

3.1.2.3 Poly(methyl methacrylate) / Acrylic	• Low production costs • Lightweight • Strong and durable • Good insulator • Will not corrode • Can be heated and shaped for flexible manufacture	• Can melt and release toxins during a fire • Not suitable to exceed temperatures of 50°C • Very difficult to recycle
	Summary: Acrylic is an inexpensive, lightweight and robust material option for the prototype construction. It will be able to insulate the subsystem components and will not corrode in the presence of water. The only flaw of this material is that it should not service at high temperatures.	

Subsequent to researching the various materials, each material was evaluated in the decision-making matrix to decide which was most suitable for the project.

Table 11: Materials Decision Making Matrix

Criteria	Material Criteria					Total
	Criteria A: Cost	Criteria C: Mass	Criteria D: Stability	Criteria E: Sustainability	Criteria F: Heat Resist	
Weighting	0.190	0.143	0.238	0.001	0.048	
Solution 1: Aluminium	4	4	4	5	5	2.529
Solution 2: Steel	3	1	5	3	5	2.146
Solution 3: Wood	5	4	1	5	1	1.813
Solution 4: Acrylic	5	5	4	3	2	2.716

It can be seen in *Table 11* above, that acrylic was considered the most appropriate material to use in unmanned firefighting vehicle. This is because the material is an inexpensive, lightweight and robust option, and is not expected to service at high temperatures as a part of the small-scale prototype. Aluminium and steel were also ranked closely behind acrylic. However, the downfall of these materials was the mass; otherwise they would have been considered the most appropriate. For this reason, aluminium and steel were considered for use sparingly in the design if necessary.

4.1.2 Estimated Bill of Materials

For the unmanned fire fighting vehicle, an estimated bill of materials for the production of the small-scale prototype was devised (*Appendix 6*). clearly shows that the construction of the prototype falls under the \$150 budget, with a total cost of \$142. This decreased cost proves beneficial to the manufacturing of the prototype as it allows parts of the budget to be left aside to replace malfunctioning components.

4.2 Technical Design Drawings

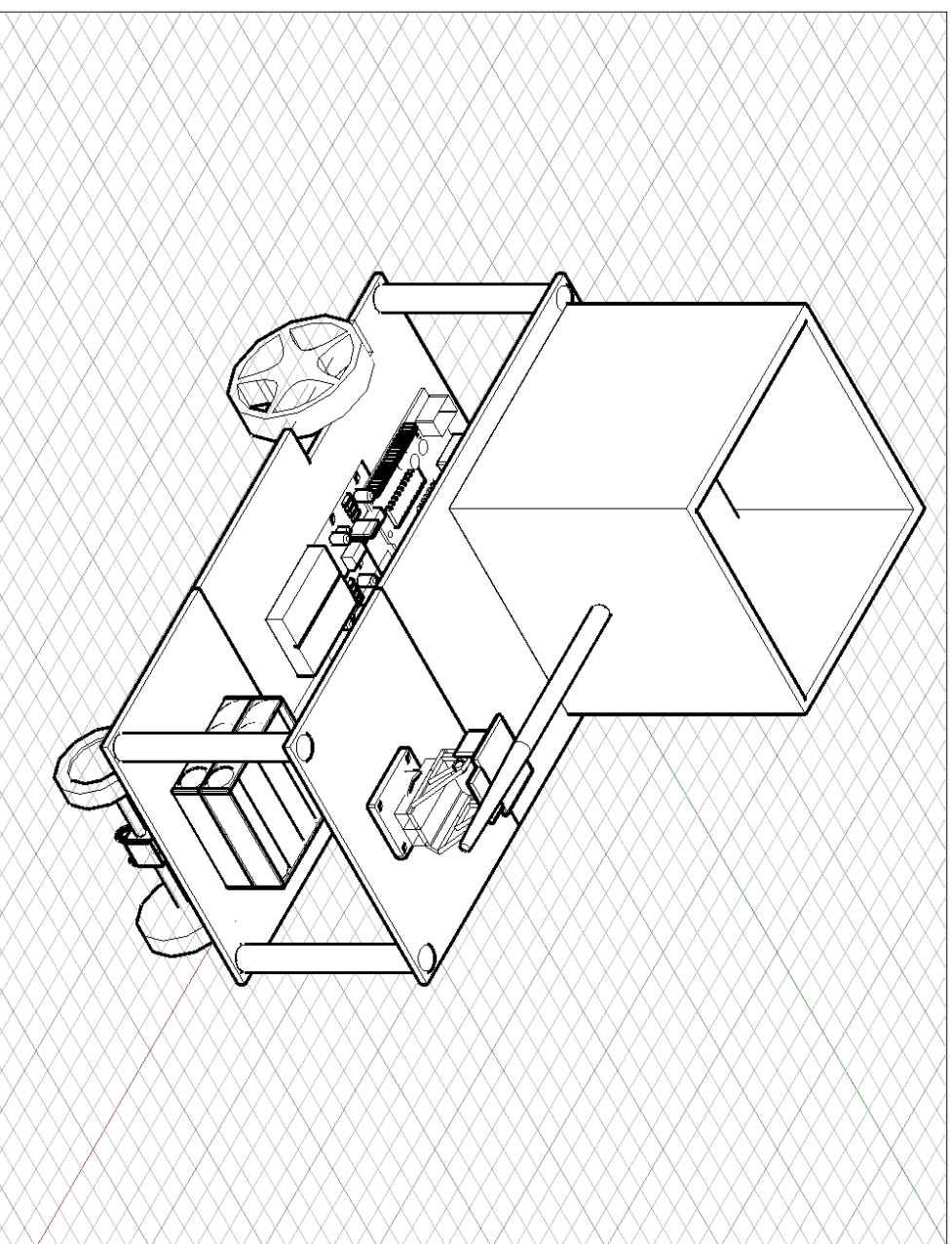


Figure 6: Technical Drawing of Final Design of Unmanned Firefighting Vehicle

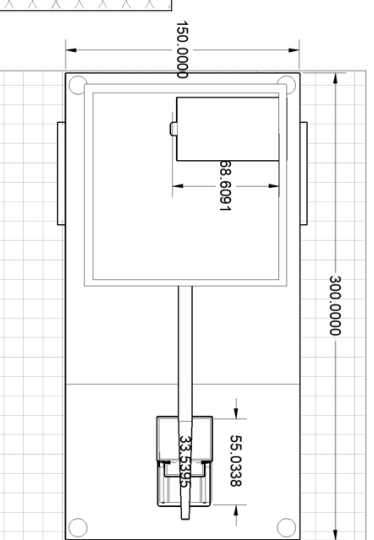


Figure 5: Top View of Final Design of Unmanned Firefighting Vehicle

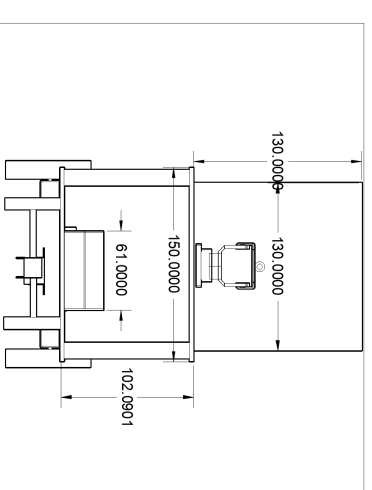


Figure 4: Front View of Final Design of Unmanned Firefighting Vehicle

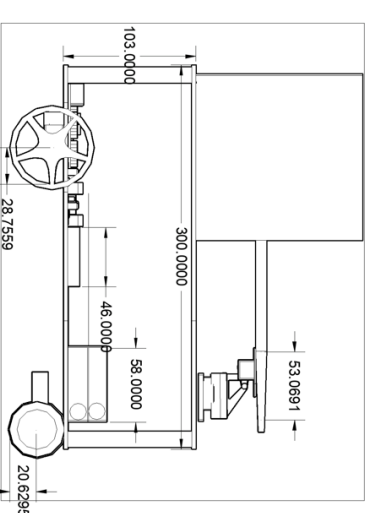


Figure 3: Side View of Final Design of Unmanned Firefighting Vehicle

4.3 Circuit Diagrams

4.3.1 Complete Circuit

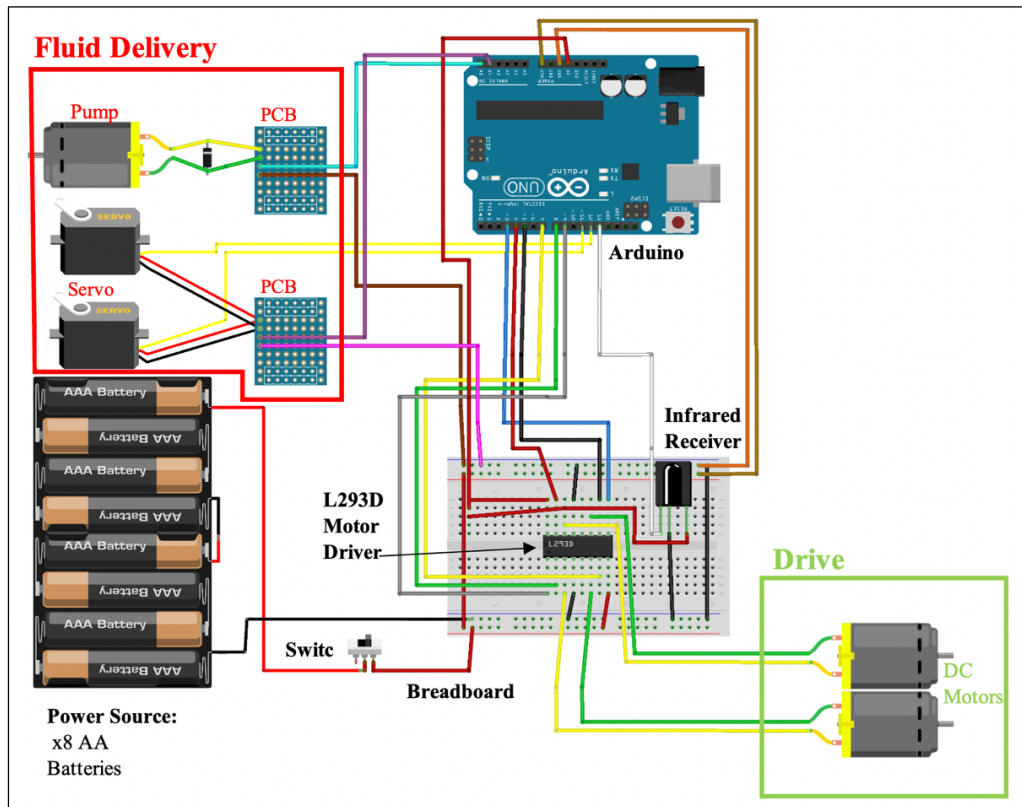


Figure 7: Full Circuit System Diagram

Figure 7 demonstrates the final circuit of the UFV prototype and how the Drive and Fluid Delivery subsystems are incorporated with the control subsystem. It also shows all the components used in the final prototype.

4.3.2 Fluid Delivery Subsystem Circuit

The circuit schematic of the PCB which is used to control the pump in the circuit can be found in *Appendix 8*. In the complete circuit diagram seen above (*Figure 7*), two separate PCBs are used; one for the pump and one for the Servo motors (*Appendix 8 & 9* where the pump and motors are connected to “J1”). This is due to the pump releasing current in the wrong direction and large amounts of EMF which affects the servo motors which require a stable voltage to function correctly. Utilising a second PCB isolates the servo motors and powers them only when required. This prevents the reverse current from affecting the servo motors. A diode was also placed in parallel with the positive terminal facing the positive terminal of the pump to redirect the reverse current.

4.3.3 Drive Subsystem Circuit

The Drive subsystem consists of 2 DC motors controlled using a L293D Motor Driver chip as seen in *Appendix 10 & 11*. The L293D chip was required as it allows the motors to be powered by external power since the Arduino does not supply enough current to control DC motors.

A motor driver can run up to 4 motors, but for the drive subsystem, was adapted to control 2 DC motors. The output pins are connected to each terminal of the DC motors and the “enable” and “input” pins are connected to the Arduino (*Appendix 10 & 11*). The enable pins are used to control the amount of power flowing through each motor and the input pins are used to control the direction of current.

4.4 Life Cycle Analysis

The aim of the Life Cycle Analysis is to assess the sustainability of the large scale unmanned firefighting vehicle. Primarily assessing environmental impacts associated with the manufacturing, use and end-of-life stages of materials and fuels used in the vehicle. The analysis is completed following the assumption that the fundamental materials and fuels used in current emergency vehicles will be the same used in the UFV. Stainless steel functions as the most common material used in apparatus bodies of fire trucks, the tank is most commonly made out of polypropylene, a durable and insulated thermoplastic and diesel fuels most emergency vehicles. The tables below show the process flow for manufacturing, use of vehicle and recycle/disposal of vehicle

Table 12: Process flow of life cycle of materials and fuels used in UFV

Production and manufacturing		
Material/Energy Inputs	Waste/emissions	References
Stainless steel (SS)	- Material offcuts - CO₂ emissions from production - Water emissions, chemicals used during production - Mass production of stainless steel in the SS industry emits air pollution - SS production incorporates high level of scrap use (50-80% recycled SS)	(ASSDA, 2020)
- Chassis frame		
- Mounts		
Polypropylene (PP)	- Plastic offcuts - Low air and water emissions	(Fiber Visions, 2021)
- Tank		
- Equipment		
Diesel	Air emissions from refining oil into gasoline	
- Fuel		
Use of robot		
Material/Energy Inputs	Waste/emissions	References
Stainless steel	Requires little maintenance and rare replacement – corrosion resistant	(ASSDA, 2020)
Polypropylene	- Requires little maintenance and rare replacement - durable and resistant to different chemical solvents, acids, and bases - Easily cleaned without high energy or environmentally unfriendly methods	(Leblanc, 2019) (Fiber Visions, 2021)
Diesel	- Air emissions from fuel used in combustion engine - CO₂ emissions increase greenhouse gases - Emit carbon monoxide, hydrocarbons and nitrogen oxides which have adverse effects to people in close proximity to exhaust	(Majewski, 2012) (Reşitoğlu, Altinişik, & Keskin, 2015)
Recycle/Disposal of robot		
Material/Energy Inputs	Waste/emissions	References
Stainless steel (SS)	50% of stainless steel is recycled into new SS products – recycling of steel uses energy - Some SS is disposed in landfill – no large impact to soil or ground water	(ASSDA, 2020)

Polypropylene	• Recyclable	(Tokiwa, Calabia,
	• Non-biodegradable (can be degraded when exposed to ultraviolet radiation from sunlight)	Ugwu, & Aiba, 2009)
	• Take 20-30 years to degrade in landfill	(Leblanc, 2019)
	• Toxic additives used in plastic such as lead and cadmium can cause adverse environmental effects	

The tables below calculate the quantitative sustainability impacts of the production, use and end-of-life stage of the full-scale vehicle, it is assumed that the weight of the vehicle is similar to a large car or medium sized truck. Table 13 displays the materials and fuels with their corresponding embodied energy, the energy used in the manufacturing process of materials.

Table 13: Embodied energy of materials/fuels in the manufacturing/production process

Material/fuel	Approx. amount (kg or L)	Embodied energy (MJ/kg) / (MJ/L)	References	Energy (MJ)
Stainless Steel	900	77	(Ahmad, 2020)	69300
Polypropylene	60	64	(Victoria University of Wellington, 2021)	3840
Diesel (L per 100km)	28.6	38.6	(Rodrigues, Martins, Nunes, & Quintas, 2018)	1103.96

The following table outlines the approximate CO₂ air emissions during the production process.

Table 14: CO₂ emissions of materials/fuels in the manufacturing/production process

Material/fuel	Approx. amount (kg or L)	Carbon emissions [per 1000kg or L] (kg)	References	Carbon Emissions (kg)
Stainless Steel	900	1900	(Bellona Europa, 2019)	1710
Polypropylene	60	275	(Heinen & Johnson, 2005)	16.5

The following table outlines the approximate CO₂ air emissions during the use of the UFV.

Table 15: CO₂ emitted during use of vehicle

Material/fuel	Approx. amount (L)	Carbon emissions [per 100L] (kg)	References	Carbon Emissions per 100km (kg)
Diesel (L per 100km)	28.6	268	(Cottingham, 2021)	76.7

The table below is the energy used in recycling the materials to be used again in either the production of the same or different materials. The stainless-steel energy is calculated under the assumption that 50% of its original weight is recycled.

Table 16: Embodied energy of materials/fuels in the recycling process

Material/fuel	Approx. amount (kg or L)	Embodied energy (MJ/kg)	References	Energy (MJ)
Stainless Steel	450	22	(Ahmad, 2020)	9900

Production and manufacturing of stainless steel utilises 50-80% recycling of pre-used stainless-steel, thus through recycling, embodied energy can be saved as less raw materials must be sourced. This energy can then be reallocated to the recycling process of stainless-steel. Although stainless steel produces 1710 kg of CO₂ emissions (see *Appendix 12*), the 50% recycling involved in the production of the material reduces the overall emissions created in its manufacturing process compared to remaking the primary material again. Furthermore, stainless steel, in comparison to alternative chassis materials such as aluminium or galvanised steel last longer as it is extremely corrosion proof, lasting up to 50 years, decreasing the need for maintenance and replacement of vehicle parts (ASSDA, 2020). As the unmanned firefighting vehicle is centrally located on the airport runway, it does not require travelling long distances, thus the use of diesel fuel will not have copious amounts of air emissions.

As the burning of diesel fuel create 76.7kg of CO₂ emissions per 100km travelled, in the future to reduce air emissions, alternative power source should be considered such as hybrid or fully electric engine systems. To reduce energy consumption in the manufacturing process, the collection and reuse of scrap materials should be prioritised. When scrap materials is re-used, it reduces the amount of raw materials needed to be mined and transported, thus reducing overall embodied energy.

4.5 Failure Mode and Effect Analysis

A Failure Modes and Effects Analysis tables was used to systematically identify potential hazards and rate their severity, probability and detectability, as well as identify possible improvements and mitigation strategies. They are used to effectively eliminate issues before they occur, which is especially important with high risk and high severity situations.

Table 17: Failure Mode and Effect Analysis

Item	Failure Mode	Causes (Failure Mechanism)	Effects	Risk Priority Rating				Recommended Improvement
				Sev	Prob	Det	RPN	
Bolts and nuts	Loose bolts and nuts holding the prototype together	Did not tighten nuts; nuts loosen while operating	Prototype becomes unstable and potentially falls apart	3	2	1	6	Fully tighten nuts using a wrench; glue or other adhesive on the nut to consolidate the hold
Circuit components	Short circuit	Poor wiring; wires become loose and touch other wires	Damage to circuit components – become unusable; hot components; explosion	4	2	2	16	Check all components are wired correctly; secure wires to their places; solder wires which can be soldered
	Wires come loose; no longer connected	Unsecured wires loosen while operating	Certain subsystems no longer functional; full system no longer functional	4	3	1	12	Secure wires to their places; solder wires which can be soldered

Water Tank	Water Leaks from Tank	Cracks and holes from poor construction	Water leaks onto circuit components – potentially damaging components; loss of water while operating	3	2	1	6	Test fill tank with water; line tank with a film or finishing spray to cover missed cracks
	Water Spills from Tank while operating	Sudden Movement of Prototype	Water leaks onto circuit components – potentially damaging components; loss of water while operating	3	4	1	12	Cover the tank with a lid or film; or refine code to reduce any sudden movements
Components	Components come loose while operating	Movement from prototype	Operation loosens friction fit of servo motor and tires and comes off	3	2	1	6	Secure tires and servo motors with glue or other adhesives

4.6 Scale-up Considerations

The objective of the task was to complete an unmanned firefighting vehicle that could extinguish remotely extinguish fires. For the full-scale task, the components would need to be exchanged for suitable larger scale use. Heat resistance will also need to be considered when designing the full-scale version as fires would be unpredictable and may damage the unmanned firefighting vehicle. When scaling up the model there are opportunities to refine the vehicle and limitations of the current vehicle that would need to be rectified, as discussed below in *Table 18* and *Table 19*:

4.6.1 Challenges

Table 18: Challenges of Scale-up

System Component	Challenges	Improvements to be made
Acrylic	- Not heat resistant. - Will require greater durability	Steel will be used in the full-scale model due to its strong thermal resistance and it's ability to support a greater mass than acrylic.
IR remote control system.	- Short range may endanger firefighters - May receive interference at airport.	A 151MHz FM series remote and receiver set will be used as it provides a much greater range and does not receive interference. (Elsma, 2021)
Batteries	- Not sustainable to scale up - Will not hold enough charge for the scaled-up vehicle	A lithium ion battery can be used as it will store a greater charge and is more sustainable as it is rechargeable. (Battery University , n.d.)
Tank	- Lacks heat resistance. - Requires greater durability	Steel can be used for the full system with an inner lining that is chemically resistant if firefighting fluid would be used.
Motors	- Lacking power to drive the full-scale model - Turning may be required in the full-scale model.	An AC synchronous motor similar to the ones used in electric cars will be used. It has a greater turning and speed control than other large-scale motors.
Pump	The pump lacks suitable power to expel a suitable volume of firefighting liquid.	An industrial firefighting pump can be used.
Arduino	An Arduino would not be suitable for the much higher voltages that would be used	Specific existing solutions that can be bought do not exist however a similar control design to existing solutions may be used to create a custom circuit board and control system for the full-scale vehicle.

4.6.2 Opportunities

Table 19: Opportunities for Scale-up

Component.	Opportunities
Fluid	Firefighting fluid can be used in place of water which can more efficiently extinguish fires.
Nozzle	A firefighting nozzle can be used which will greatly increase the efficiency of the firefighting fluid. (Sasaki, Miyazato, Inamoto, Yamamoto, & Nakamura, 2015) the nozzle can also be adjustable in spread to suit the situation.

Control	Additional control options can be added in the coding to give greater adaptability of situations. The control subsystem can be stored in a heat and water resistant casing to reduce the risk of the subsystem failing.
Servomotors	The servomotor system has the greatest accuracy for positional control and speed however the same design in the full-scale model may expose the motors to heat damage. A geared system in conjunction with the servomotors would be a suitable system.
Structure	Metal bolts and welding can be used to affix the components together securely. The lower level of the vehicle can be closed so that the internal components will not be damaged.

5. Reflection

During the testing process, it was necessary to test all components individually. This aimed to identify any possibility for failure, so that replacement components could be sourced in a timely manner. During the drive subsystem testing, an error was made that resulted in too much power being sent through the motor, causing the gears to shred and wheels to break. Resultantly, replacement gears, axels and wheels were purchased and tested once again with a lower power. Taking such precautions and learning from errors made easily resolved the issue, and preventative measures were taken to ensure a similar situation would not occur.

While in the coding process there were a few issues however they were all resolved. The first issue we experienced was while testing the pump it only turned on for a short amount of time before turning off. We knew this issue was with the PCB as its purpose is to increase the current to allow the pump to run better. We knew how to integrate it however we were worried that the current would be too much since the moto shield is doing the same as the PCB. We used a transistor as there appeared to be a clash with the code libraries which are required to run to DC motors and servomechanisms. This impacted the fluid delivery subsystem, as if this issue was not resolved then the rotation used to aim the nozzle would not be suitable. These issues were resolved from online research as well as help from others. Our mistakes were clear in this situation, and we knew what was needed to resolve it if a similar situation was to occur.

Once we begun testing the fluid delivery subsystem with the coding completed, we had originally utilized zip ties in order to attach the nozzle to the servomechanisms. However, after testing it was clear that the zip ties were interfering with the motor. This disrupted the rotation of the servomechanisms and they needed to be replaced. We then utilized glue to hold the nozzle in place, which worked effectively. From this experience it was clear that it was more effective to use glue rather than an object that could potentially interfere with another component and cause greater issues.

The final task we faced during the testing process of the vehicle was an exposure of water on the PCB and Arduino. This occurred from testing the prototype without a lid, as the system came to a sudden stop and the water spilt out of the container. This meant that we had to let the PCB and Arduino dry out for a day before continuing to use them, so this took away a day of testing time. After this situation we utilized the lid of the container which was effective, and no further issues occurred.

On the day of demonstration our prototype performed a perfect score for the quality test as our system moved freely with a stable structure and a well working control system with a controlled spray system. However, we only scored 13.59 out of 60 for our performance. This was a result of only extinguishing one fire and wasting water when not hitting the targets, which resulted in an inefficient water usage. Overall, our system scored 50.59% for the total performance. This score could have been improved greatly with a smoother rotation for the aiming system, as the main issues was aiming accurately. From this experience it is clear that with a better controlled rotation of the nozzle which could include a different type of remote, this issue would be resolved.

6. Conclusion

The Australian Aviation Rescue Fire Fighting Service commissioned the development of an unmanned firefighting vehicle, which was required to have a robust structure must contain the drive, fluid delivery and control mechanisms, and must resist movement during operation.

The proposed design satisfies the design criteria with intuitive control mechanisms combined and precision-based fluid delivery system. The unmanned fire fighting vehicle was able to efficiently move to and subsequently extinguish targeted fires. Due to the vehicles low-cost design, a scale up and mass deployment of unmanned fire fighting vehicles is not only possible but also recommended.

6.1 Recommendations

There are a number of constraints of this investigation, which are recommended for further study. The small-scale unmanned fire fighting vehicle was a suitable proof of concept, but the design is not ready for a fully scaled up vehicle, as was proven in the demonstration. *Table 20* showcases the recommendations for the small-scale prototype before a full-scale vehicle can be developed.

Table 20: Project Recommendations

Type of Project	Project Definition	Reason Behind Project
Research	Investigation into other structure-based materials that would prove to be equally lightweight but have a higher structural load tolerance and heat resistance.	The vehicle would be expected to perform its tasks in non-ideal environments where it would be exposed to the unpredictable nature of fires and also the increased structural load of the full-scale vehicle.
	Investigation into alternate energy sources that prove to be more energy dense and less heavy compared to the AA batteries used in the prototype.	The AA batteries used in the prototype would be far too impractical in a full-scale vehicle and also, the weight of the AA batteries proved to be more than acceptable in the small-scale prototype.
Development	Further environmental protection to secure the vehicle from hazards both internal and external.	Concerns were raised surrounding the more open design of the vehicle which would expose sensitive components to the heat of a fire and also to the water stored on the vehicle.
	Experimentation surrounding different forms of nozzles to determine idealised flow rate.	Minor concerns surrounding the flow rate were present. An experiment to determine the idealised flow rate would allow a more efficient firefighting experience.
	Experimentation of different servo motors for the aiming of the fluid delivery is required to provide a smoother aiming experience.	The servos used proved to be extremely unreliable, often breaking for no discernible reason and they proved themselves to be inefficient when it came to the movement of the mount.

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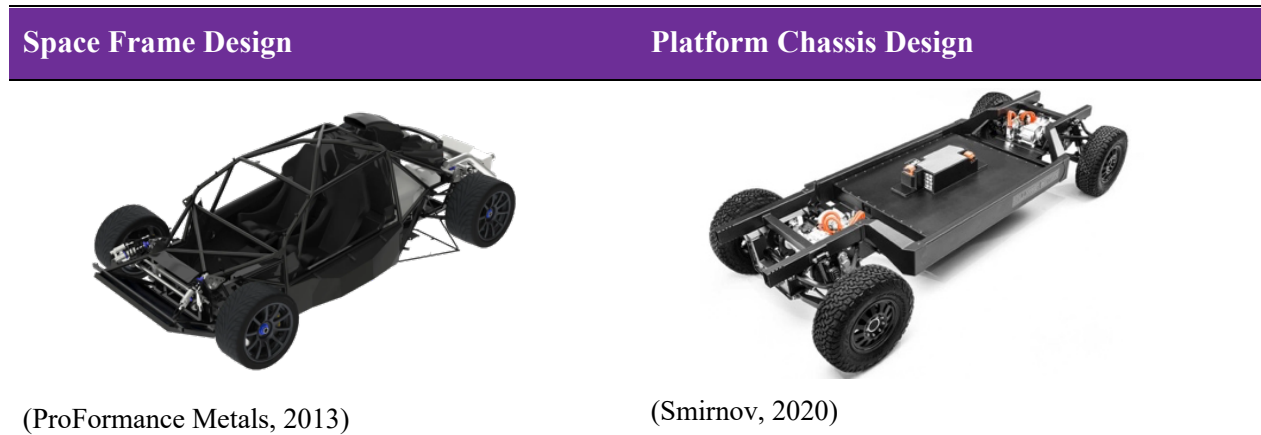
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8. Appendix

8.1 Appendix 1: Structural Subsystem Existing Design Inspiration



8.2 Appendix 2: Structural Subsystem Pairwise Comparison Matrix

Criteria	System Criteria			Subsystem Criteria				Sum Total	Weighting
	Criteria A: Cost	Criteria B: Size & Form	Criteria C: Sustainability	Criteria D: Durability	Criteria E: Heat Resist	Criteria F: Mass	Criteria G: Repairability		
Criteria A: Cost		B	A	D	A	A	A	4	0.190
Criteria B: Size & Form			B	B	B	B	B	6	0.285
Criteria C: Sustainability				D	E	F	G	0	0.001
Criteria D: Durability					D	D	D	5	0.238
Criteria E: Heat Resist						F	G	1	0.048
Criteria F: Mass							F	3	0.143
Criteria G: Repairability								2	0.095

8.3 Appendix 3: Fluid Delivery Subsystem Pairwise Comparison Matrix

Criteria	System Criteria			Subsystem Criteria			Sum Total	Weighting
	Criteria A: Cost	Criteria B: Size & Form	Criteria C: Mass	Criteria D: Water Efficiency	Criteria E: Rotation of Nozzle	Criteria F: Water Pressure		
Criteria A: Cost		A	A	D	E	E	2	0.13
Criteria B: Size & Form			C	B	B	B	3	0.20
Criteria C: Mass				D	E	F	1	0.07
Criteria D: Water Efficiency					E	D	3	0.20
Criteria E: Rotation of Nozzle						E	5	0.33
Criteria F: Water Pressure							1	0.07

8.4 Appendix 4: Control Subsystem Pairwise Comparison Matrix

Criteria	System Criteria			Subsystem Criteria			Sum Total	Weighting
	Criteria A: Cost	Criteria B: Size & Form	Criteria C: Mass	Criteria D: Voltage/Current	Criteria E: Accessibility	Criteria F: Longevity		
Criteria A: Cost		B	C	D	A	F	1	0.0667
Criteria B: Size & Form			B	D	B	B	4	0.2667
Criteria C: Mass				D	E	F	1	0.0667
Criteria D: Voltage/Current					D	F	4	0.2667
Criteria E: Accessibility						F	1	0.0667
Criteria F: Longevity							4	0.2667

8.5 Appendix 5: Drive Subsystem Pairwise Comparison Matrix

Criteria	System Criteria			Subsystem Criteria				Sum Total	Weighting
	Criteria A: Cost	Criteria B: Size & Form	Criteria C: Mass	Criteria D: Durability	Criteria E: Sustainability	Criteria F: Power	Criteria G: Movement		
Criteria A: Cost		A	A	D	A	F	G	3	0.14
Criteria B: Size & Form			C	D	E	F	B	1	0.05
Criteria C: Mass				D	C	F	C	3	0.14
Criteria D: Durability					D	D	G	5	0.24
Criteria E: Sustainability						F	G	1	0.05
Criteria F: Power							G	4	0.19
Criteria G: Movement								4	0.19

8.6 Appendix 6: Calculation of Bending Deflection of Acrylic

$$\delta_{MAX} = \frac{PL^3}{48EI}$$

$$\delta_{MAX} = \frac{PL^3}{48E \left(\frac{Lh^3}{12} \right)}$$

$$\delta_{MAX} = 1.7723 \text{ mm}$$

$$\text{where: } P = 9.8 \text{ N}$$

$$L = 250 \text{ mm}$$

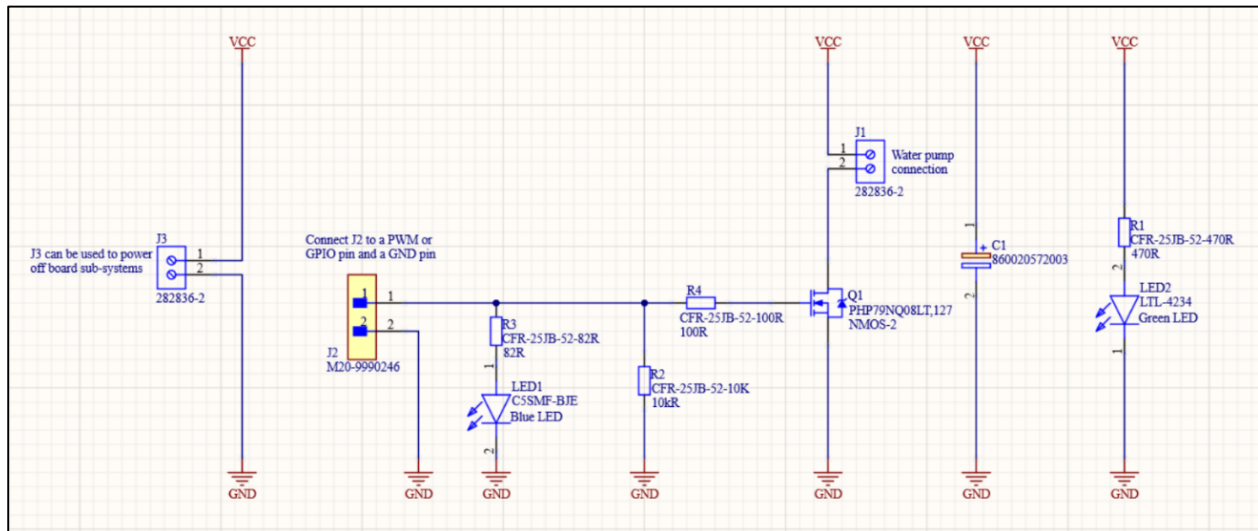
$$h = 3 \text{ mm}$$

$$E = 3 \text{ 200 MPa}$$

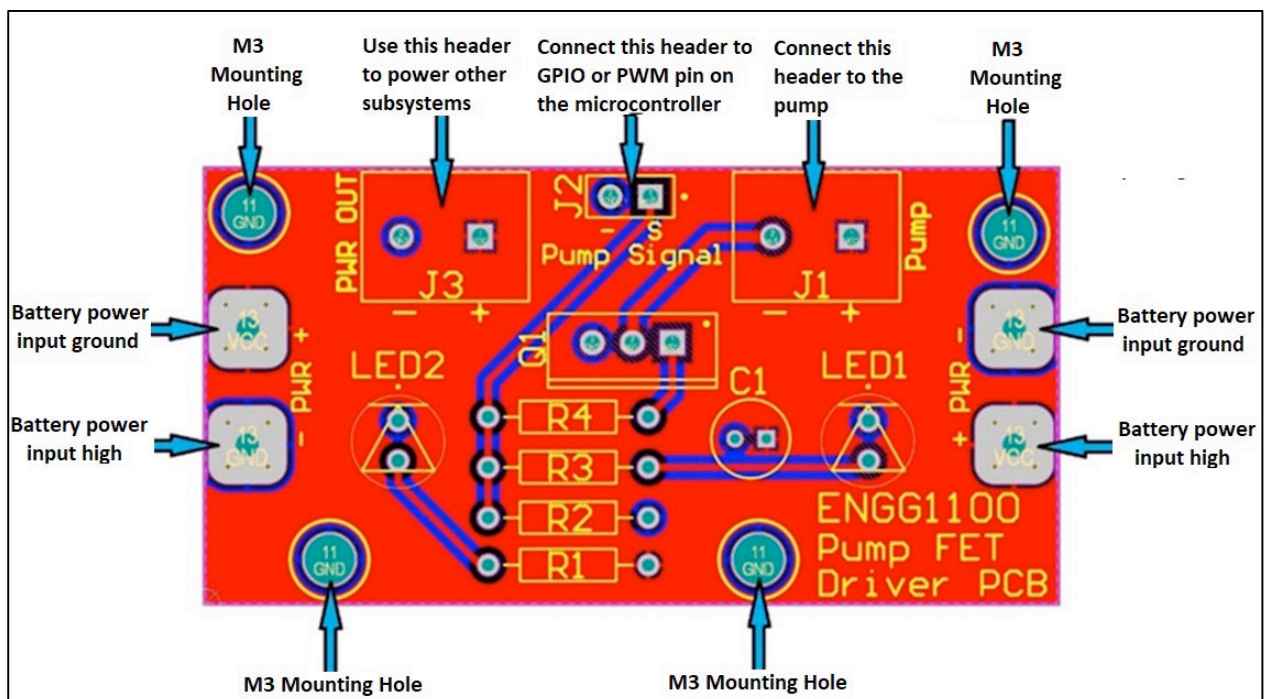
8.7 Appendix 7: Bill of Materials

Code	Budget Description	Total Cost (\$)	Cost per Unit Used (\$)
PB8817	Breadboard	4.95	4.95
SB21334	AA Alkaline Battery	7.95	5.3
XC4618	Servo Bracket Mount	4.95	4.95
WC6027	Jumper Lead 100pc	14.95	2.99
YM2758	Micro Servo Motor	11.95	23.9
XC4472	Servo Control Motor	12.95	12.95
PH9209	8 x AA Battery Holder	2.95	2.95
19800530	Threaded Aluminium Rod	5.95	4.4625
9370048217	Hex Half Nut	4.49	2.39
9370048243	Machine Washer	5.99	2.13
WP01720940	3mm Acrylic	12.99	6.495
P_42972051	1L Food Container	5	5
CE05345	Mini Submersible Pump	3.25	3.25
YG2900	Geared Motor w/ Wheel	9.95	19.9
YG2742	Universal Gearbox Motor	22.95	11.475
N/A	Arduino Uno	20	20
N/A	Arduino Base Plate	5	5
ST0355	Toggle Switch	3.75	3.75
Total Expenditure (\$)			141.8425

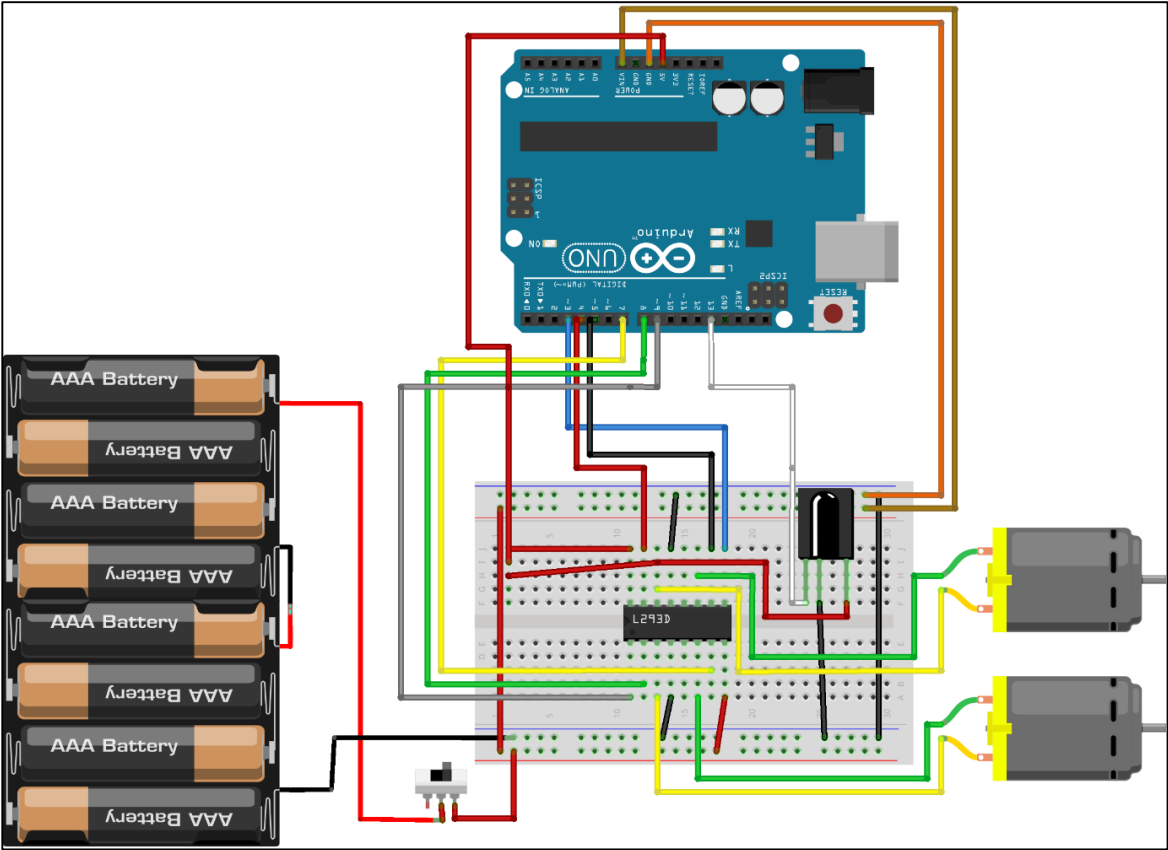
8.8 Appendix 8: PCB Circuit Schematic



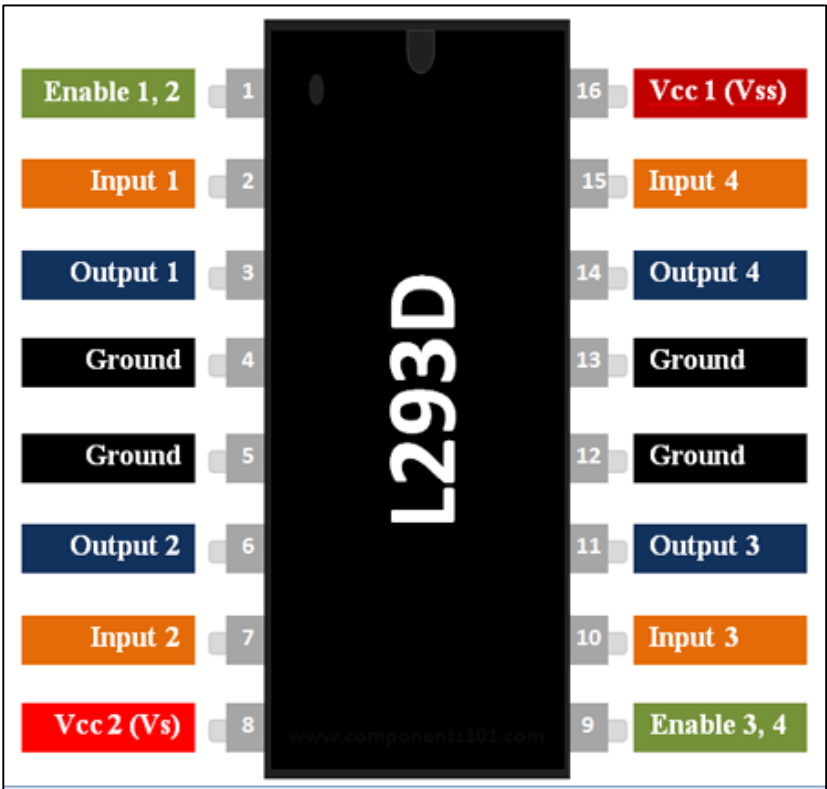
8.9 Appendix 9: PCB Diagram



8.10 Appendix 10: Drive Subsystem Circuit Diagram



8.11 Appendix 11: L293D Motor Driver Pin Layout



8.12 Appendix 12: Embodied energy of materials/fuels in the recycling process

Material	Reasoning/evidence for mass/volume (L)
Stainless steel	The average SUV or car uses approximately 900kg of steel (Maverick, 2020)
Polypropylene	In assuming that the scaled-up firefighting vehicle will be similar in size to a middle sized firefighting vehicle, their water tanks generally hold 2000 litres (CFA, 2021). A polypropylene tank of that size weighs approximately 60kg (National Poly, 2021)
Diesel	Assuming that the Unmanned Firefighting vehicle would be similar to a 'rigid truck' in consuming fuel, the average fuel consumption would be approximately 28.6 L per 100kms. (Budget Direct, 2020)